Hanxi (Gary) Chen

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EDUCATION

Cornell University — Ithaca, NY
Ph.D. in Computer Science, Programming Languages — Present
Advisor: Prof. Alexandra Silva, Prof. Andrew C. Myers

University of Pennsylvania, *Penn Engineering & The Wharton School* — Philadelphia, PA
M.S.E. in Computer and Information Science — May 2025
B.S.E. in Computer Science, Mathematics minor
B.S. in Economics
Cumulative GPA: 3.97/4.00, *Summa Cum Laude*
Advisor: Prof. Steve Zdancewic, Prof. Sanjeev Khanna

SELECTED PUBLICATIONS

Hanxi Chen, Noam Zilberstein, Andrew C. Myers, Alexandra Silva. Oblivious Probabilistic Outcome Logic: Verifying Probabilistic Programs with an Oblivious Adversary. arXiv, 2026.

Calvin Beck, Hanxi Chen, Steve Zdancewic. Vellvm: Formalizing the Informal. The Eleventh International Workshop on Coq for Programming Languages (CoqPL), 2025.

Calvin Beck, Hanxi Chen, Steve Zdancewic. Vellvm: Formalizing the Informal LLVM (An Experience Report). NASA Formal Methods: 17th International Symposium (NFM), 2025.

Calvin Beck, Irene Yoon, Hanxi Chen, Yannick Zakowski, Steve Zdancewic. A Two-Phase Infinite/Finite Low-Level Memory Model: Reconciling Integer–Pointer Casts, Finite Space, and undef at the LLVM IR Level of Abstraction. Proceedings of the ACM on Programming Languages, 8 (ICFP), 2024.

RELEVANT RESEARCH EXPERIENCE

Research Assistant, *Cornell PL Group*, supervised by Prof. Alexandra Silva and Prof. Andrew C. Myers — Sep 2025 – Present
Cornell University, Computer Science

- Developing program logics for verifying probabilistic programs under oblivious schedulers, with applications to randomized distributed protocols and hardware.
- Building a Lean mechanization library for probabilistic reasoning, including probability-space constructions, outcome triples, and proof automation.

Research Assistant, *PLClub*, supervised by Prof. Steve Zdancewic — Jun 2021 – Aug 2025
University of Pennsylvania, Computer and Information Science

- Contributed to Vellvm in Steve Zdancewic's group, formalizing LLVM IR semantics and low-level memory models in Rocq.
- Built Rocq/OCaml differential-testing tools for validating Vellvm's semantics and finding bugs in LLVM implementations.

Research Assistant, *Wharton Research Scholars*, supervised by Prof. Sanjeev Khanna — Jan 2023 – Dec 2024
University of Pennsylvania, The Wharton School

- Studied precedence-constrained knapsack problems, proving pseudo-polynomial algorithms for structured graph classes.
- Wrote a 48-page undergraduate technical report and presented the work in the Wharton Research Scholars seminar.

TECHNICAL SKILLS

Proof Assistants: Lean, Rocq/Coq, Mathlib, QuickChick
Programming: OCaml, Haskell, C/C++, Python, SQL, SystemVerilog
Systems and Tools: LLVM IR, Clang, Dune, Git, Unix, LaTeX
Research Methods: mechanized semantics, program logics, differential testing, property-based testing, randomized algorithms